This is an excerpt from Frank Elavsky's dissertation on *Tool-making as an Intervention on the Accessibility of Interactive Data Experiences*, which can be accessed in full at this archival link:

http://reports-archive.adm.cs.cmu.edu/anon/hcii/CMU-HCII-26-103.pdf

This document contains the following sections:

- Abstract
- Chapter 1: Introduction (Section 1.1 only)
- Chapter 2: Background & Related Work
- Chapter 3: Overview of Contributions
- Chapter 9: Findings (Sections 9.1, 9.2, and 9.6 only)
- Chapter 10: Discussion & Future Work (Sections 10.2 and 10.3 only)
- References

This document does not contain the following chapters:

- Chapter 4: *Chartability*: Heuristics as a Tool and Resource
- Chapter 5: *Data Navigator*: Low-level Tooling for Creating Navigable Data Structures
- Chapter 6: *Skeleton*: Visual Authoring of Non-visual Data Experiences
- Chapter 7: *Cross-perception*: Rethinking Input Design Towards Blind Analytical Interaction
- Chapter 8: *Softerware*: Enabling Personalization of Interactive Data Representations for Users with Disabilities
- Chapter 12: The *Softerware* Option Taxonomy
- Chapter 11: Biographical Sketch

Abstract

In this dissertation, we contribute practical advancements in tool-making as an intervention on the accessibility of interactive data experiences. The thesis of this dissertation is as follows:

This dissertation argues that the tools practitioners use to build interactive data experiences are themselves sites where accessibility barriers are produced, prevented, or alleviated for both end users and authors. This work contributes five tools, Chartability, Data Navigator, Softerware, Cross-perception, and Skeleton, that collectively center accessibility work on the empowerment of disabled and non-disabled practitioners across the full arc of evaluation, data navigation, analytical interaction, and personalization.

Rather than framing accessibility research solely around ideal experiences for end users with disabilities, this thesis investigates why accessibility work is so difficult for the practitioners who build interactive data experiences and what tool-making can reveal about those difficulties. We organize this investigation around four domains where practitioners face the most persistent challenges: evaluation, navigation, interaction, and personalization. *Chartability*, a heuristic framework contributed first and maps the full landscape of accessibility barriers. *Chartability* helped us to identify and prioritize our three latter domains (navigation, interaction, and personalization) as the areas where the most severe gaps remain in practitioner work. *Data Navigator* and *Skeleton* then investigate navigation, finding that practitioners struggle because navigation structure has no visible, manipulable representation in their workflows. *Cross-perception* engages interaction, demonstrating that blind data analysis has been constrained by existing tools and that a new interaction design framework can reshape what analytical work is possible. *Softerware* addresses personalization, revealing that access needs genuinely conflict across users and that meaningful personalization requires system-level infrastructure that does not yet exist in practitioner tooling ecosytems.

Combined, these contributions provide empirical insights and practical advancements in the state of the art for tooling that bridges gaps in current accessibility practices in visualization and data science. Our work ultimately enables people with and without disabilities to better evaluate barriers in, analyze with, design for, develop, and personalize interactive data experiences. We demonstrate that tool-making is a productive intervention that both engages accessibility barriers and elucidates why those gaps exist in practitioner work.

Part I

Introduction

Chapter 1
Introduction

This thesis is a body of research situated within the existing research area focused on making interactive data representations more accessible for people with disabilities. Much of the work in this existing area is situated within the context of making interactive data *visualizations* accessible, particularly (but not exclusively) for people who are blind. Our work, contributed here in this thesis, is focused on using *tools* as a specific intervention and sub-area of study for making interactive data *representations* accessible for people with disabilities, broadly speaking. ("Representations" here is an intentionally broader term than "visualizations," which are exclusively visual representations of data.)

Before we begin, two things must be understood up front, or else the rest of this thesis could be interpreted with disruptive assumptions: we must interrogate the phrase "making visualizations accessible" and unpack why *tools* are a meaningful area of study.

1.1 Is "accessible visualization" really an oxymoron?

The first assumption that must be disrupted is perhaps the motivating cornerstone of this research, which is that the phrase "making visualizations accessible," while a noble goal, is not the semantically correct phrasing nor precisely what describes our work. This can be misleading. We do use the phrase "accessible visualization" but will admit that this seems to confuse certain people with very particular opinions about things. We will clear this up.

Villains of our field's past have written incendiary and ableist perspectives on why "no forms of data visualization, not just dashboards jam-packed with graphics, can be made fully accessible to someone who is blind," and that "[a blind man] will never be able to analyze data as I do visually, because many aspects of vision cannot be duplicated by his other senses" [30]. However, this position misunderstands what the goal of accessibility is, and arguably even what the goal of visualization itself is.

Making visualizations accessible *isn't* about the visualization, it's about making the outcomes of the visualization accessible.

Visualizations are ubiquitous and paramount for decision-making. However, the *artifact* that is a visualization is not even the goal of the act of visualizing: developing understanding, insight, confidence, and communication among and between human beings are the goals of visualization. Visualization is about making data easier to use for all kinds of things. Yes, our visual system enables us more than any other form of sensory cognition that we have [18, 36, 115]. But we aren't trying to make sight itself accessible. We are trying to make it possible for people to make meaningful decisions, gain valuable information, build conjectures, and effectively communicate with others.

Many, many people who I, Frank, have spoken to over the course of my career, even before embarking on this thesis journey, misunderstand this simple fact: making a "visualization" accessible *isn't* about the visualization itself but rather making what the visualization is meant to *accomplish* accessible. It's about equal outcomes, not equal interactions with an artifact.

People with disabilities are no small portion of the world's population. In the United States, 27% of people self-report living with at least one disability that affects their daily lives [92] and all of us will eventually age into disability (if we are lucky to live a long life).

People with disabilities (again, that will be all of us *eventually*) deserve to participate fully in life. They deserve financial independence. They also deserve loving care and interdependence [8, 57, 88]. People with disabilities have a right to make informed decisions, to know about the status of a global pandemic, and to have an understanding of local and national politics [29]. While we use visualizations to navigate all of these domains, the goal is not to make the charts and graphs themselves somehow equally useful to all people. That would be a false measurement of success.

Our goal then, is measured by the success of lives led by people with disabilities [120]. Many other measurements are just metrics along the journey towards that goal. We then ask: Can people with disabilities also use data to live full lives? Can they make *fast* decisions based on data? Meaningful, careful, *slow* decisions? Communicate complex ideas? Crunch and clean data, develop models, find errors, and build hypotheses? Can they have memorable, immersive, beautiful, aesthetic experiences with data too [57]? Ultimately, our aim is not accessible "visualizations" but equitable and accessible *outcomes*, everything an interactive data experience *accomplishes* for the people who use it.

Again, if the goal of accessible visualization were about visualizations themselves, then the correct course of action would be one framed by the medical model [82]: that there is a normative state of behavior and capability (in this case, it would be "normal" to be able to read a visualization and make a decision) and any deviation from that norm must be corrected. This framing first assumes that the visualization should not be altered or improved. And then this framing puts the burden on the bodies of people with disabilities: that they must be "fixed" and given sight or brought to some equivalent state as someone who is "healthy," normal, and sighted. Plenty of scholars have already discussed why this framing is a problem, not only because it places undue burden on people with disabilities, produces pathologies and hierarchies of disability, but also because it is fundamentally not economically or ethically feasible.

So we then turn to other models of disability, such as the social model. The social model is heavily discussed by disability scholars and is not the end-game or last and total way of thinking about disability [82, 95, 96, 113, 136]. But the core motivation is that society, not medicine, is also a path towards solving problems that people with disabilities face. A few important concepts and concretely actionable things come from the social model that can help motivate the work of this thesis.

First, we look to the historical birth of the social model of disability: in the 504 sit-ins that took place in the United States in 1977. Cities had curbs and curbs are a barrier for people who use wheelchairs. So protests happened because decisions were being made without people with disabilities at the table. In this instance, people acknowledged that political power was an exclusive club and fought to ensure their cry "nothing about us, without us!" materialized.

And this leads us to the first and most-foundational philosophical framing for this thesis: that our *artifacts*, these things we've created from curbs to data visualizations, can become *barriers* for people with disabilities. And it is then the artifact, not the body of the person with a disability, where disability is produced in this model. Rather than a comparison to a normative state as a way to frame disability (the medical model), we instead must observe and evaluate material outcomes based on human-made problems.

So, the social model is framed around society "solving" inequities: we get involved and make political and legal change tangible. But a second model also emerges from within the social model: one where we can now frame *who is first responsible* for repair: the curb designers and implementers.

And knowing who is first responsible for access leads us into the moral and ethical imperative that motivates this thesis: the builders and makers of visualizations are ultimately the ones who provide exclusive value for only a subset of people: those *without* disabilities. **We must first change how builders and makers do their work.**

So the phrase "accessible visualization" is really about recognizing that visualizations produce barriers for people. That means that it is our ethical imperative, as builders and makers, to fix them. And that act of fixing barriers leads us away from mere visual representations of data into a wide variety of other senses and interaction modalities. There are many paths forward towards fuller and more-equitable lives led by people with disabilities.

Chapter 2
Background & Related Work

This chapter sets up the story that the rest of this thesis engages. It makes three moves. First, we ground the idea that tools are not neutral instruments: tools shape the work and the workers who use them, which is why tool-making is a meaningful site for accessibility intervention. Second, we survey the long conversation between accessibility and data visualization and its history across modalities, including engaging critical voices that intersection with visualization and accessibility work which inform, support, and hold in tension the work of this thesis. Lastly, we argue that the hardest remaining challenges cluster in three domains, navigation, interaction, and personalization, and we signpost which chapters engage each of these.

2.1 Tools and the People Who Use Them

2.1.1 Studying Practitioners and Their Work

Human-computer interaction has a long tradition of studying the people who make things. Ethnographic and case-study methods immerse researchers in maker, DIY, and assistive technology communities to understand how practitioners actually work, especially when the work is new and unfamiliar to them [56, 58, 59]. Participatory design and co-creation bring practitioners and end users directly into the research process, surfacing how designers shift their thinking when they encounter novel problems [43, 100, 114]. And a growing body of design-based research aims not merely to fill gaps in existing practice but to extend what practitioners are capable of: scaffolding experimentation, supporting learning, and amplifying creative and technical expression [22, 41, 62, 67, 122].

This thesis sits inside that tradition but takes a particular methodological stance: we study practitioners *through their tools*. Each project in this thesis builds a tool, places it in front of working practitioners, and observes what the tool reveals about why accessibility work is hard for them. Sometimes the tool is a resource practitioners apply in their real environments, sometimes a system they build with, and sometimes a probe whose main purpose is to make an invisible part of their work observable. Tool-making here is both an intervention and an instrument of inquiry. That stance only makes sense, however, if tools really do shape work in a deep way. So before surveying the accessibility and visualization literature, we want to engage that claim seriously.

2.1.2 How Tools Use Us

We tend to talk about using tools. We talk much less about how tools use us. A chair is a tool for sitting, but a lifetime of chairs trains a body into the chair's posture. A pickaxe is a tool for extracting ore, but the mine organizes the miner: their hours, their movements, their risks, and what counts as a day's work. Tools come with a direction already built in, and the people who take them up inherit that direction.

Sara Ahmed gives this intuition theoretical teeth. In her phenomenology of orientation, objects are not passive things waiting to be used: they direct bodies, making some actions feel near and natural while other actions fall out of reach, and the social world is full of what she calls straightening devices, arrangements that keep bodies aligned with the paths already laid down for them [1]. In her later work she traces how *use* itself is formative: things become shaped by their use, "fit" for some users and not others, and a history of use leaves a trace that directs how a thing will be used next [2]. Bowker and Star make a parallel argument at the scale of infrastructure: the classification systems and standards embedded in our information systems are largely invisible scaffolding, and every standard valorizes some point of view while silencing another [16]. Assumptions about ability are among the assumptions most commonly baked in: tools encode what their makers believe people are able to do [133].

Visualization research has begun to examine its own tools in these terms. McNutt's survey of 57 JSON-style domain-specific languages for visualization shows that each grammar embodies contested decisions about who its user is and what that user may express, with recurring tensions between formal and colloquial specification and between competing user types [85]. His dissertation develops the larger argument: the design of visualization languages, and of the tools that surround them, directly shapes and can either enhance or constrain end-user agency, motivating systems that offer assistance while respecting the user's intent [86]. In other words, the field's own instruments are straightening devices. Every charting grammar that assumes a sighted reader, every renderer that emits unstructured pixels, quietly directs practitioners away from accessible outcomes, not by forbidding them but by making them feel out of reach.

Tool users are not powerless in this relationship. A rich literature on appropriation documents how people adapt tools beyond their designers' intent, and how that emergent use can be a source of design knowledge in its own right [26, 27, 99, 119], with some work building general theory from the study of emergent and generative tool use [7, 10]. HCI's systems community has likewise developed methods for building and studying tools deliberately: piggybacking on existing systems to study an improvement in context [39, 53], and contributing toolkits whose value is argued through demonstration, usage, technical performance, or heuristics. Ledo et al., analyzing 68 toolkit papers, found these four evaluation strategies dominate, with demonstration by far the most common [72]. That finding will matter later in this chapter: a field can produce many demonstrated toolkits while producing very little maintained infrastructure.

The takeaway from this section is the premise the whole thesis rests on: tools direct work. If the tools of data visualization currently straighten practitioners toward inaccessible outcomes, then re-orienting those tools is not a cosmetic fix. It is an intervention at the place where practice is formed. The rest of this chapter examines what the field has learned about accessible data experiences, and why that learning has so far failed to re-orient the tools.

2.2 The Conversation Between Accessibility and Data Visualization

Accessibility and data representation have been in conversation far longer than the modern visualization community has existed, and that conversation has two registers. The first is a history of techniques: a steady accumulation of ways to carry data to people through senses and devices other than a sighted eye on a screen. The second is a critical register, largely led by disabled scholars and disability studies, that questions how that technical work is framed, who leads it, and who it serves. This thesis tries to take both seriously, so we survey them in turn.

2.2.1 A Short History Across Modalities

Tactile representation is the oldest thread. Tactile maps and graphics for blind readers date to at least the 1830s [42], and tactile sensory substitution for charts has been studied in the computational sciences since the early 1980s [105]. Practice in this modality is unusually mature: tactile and braille standards are robust and well-maintained [6], although their assumptions (embossed paper, fixed layouts) transfer only partially to digital, interactive contexts. Research has worked on that seam, for example by augmenting tactile graphics with audio annotations [5], and industry accessibility teams have published candid experience reports on what it takes to produce tactile graphics inside a real design workflow [24]. Recent work continues to refine the perceptual foundations of non-visual structure [87].

Sonification has a similarly long arc, and because it is a substantial and active research community in its own right, it deserves explicit treatment here. Mansur et al.'s sound graphs demonstrated in 1985 that line data mapped to pitch over time could communicate trends to blind listeners [81]. Subsequent decades built out the empirical base: cross-modal comparisons of auditory and visual graphs [32], non-visual presentation of structured information [17], techniques for scanning and comparing multiple data series by ear [84], and interactive sonification of geo-referenced data [137]. The community consolidated its methods and theory in the Sonification Handbook [52]; Walker and Nees's theory chapter formalizes sonification as a mapping problem from data dimensions to auditory dimensions, emphasizes context cues that play the role axis ticks play in vision, and stresses that interaction, not just playback, shapes what listeners can extract [126]. Sonification research has also been a site of methodological progress, including co-design with blind and low vision collaborators [23].

What is most relevant for this thesis is the recent trajectory: sonification is increasingly integrated into multimodal, interactive systems rather than offered as a standalone substitution. VoxLens adds summaries, sonification, and voice queries to web visualizations through a JavaScript plug-in, and in a study with 21 screen reader users improved information-extraction accuracy by 122% and interaction time by 36% over conventional interaction [109]. Chart Reader, co-designed with screen reader users, combines structured navigation with sonification and author-configurable data insights [121]. Auditory maps have reached usable maturity for wayfinding and choropleth reading [12]. Notably, sonification is also one of the few accessible-visualization techniques to cross into production: Apple shipped Audio Graphs in 2021 as a VoiceOver feature with a public developer API [4]. And Umwelt re-frames the relationship entirely, treating sonification, textual structure, and visualization as coequal representations derived from a shared abstract model in an accessible authoring environment [140]. Sonification's trajectory, from output substitution toward interactive structure and authoring, previews the argument this chapter is building.

Textual description is the most widely deployed modality, and research has moved it well past "write some alt text." Lundgard and Satyanarayan's model of semantic content distinguishes the levels of meaning a description can carry, and shows that blind and sighted readers value different levels [74]. Jung et al. offer evidence-based guidance on description length, plain language, and the ordering of information [65]. Kim et al. collected the questions screen reader users actually ask of visualizations, pointing toward more dialogic, query-driven access [69], and Chundury et al. deepen our understanding of how sensory substitution choices should be grounded in users' practices [21].

Navigation and screen reader interaction is the youngest thread and the one closest to this thesis. Foundational work made statistical and mathematical content traversable by screen reader [37, 112]. Studies of screen reader users' experiences with two-dimensional digital content, including visualizations, documented both the depth of exclusion and the strategies users develop anyway [103, 108]. Zong et al. then gave the area its clearest design vocabulary: structure (how a visualization's entities are organized for traversal), navigation (the operations that move through that structure), and description (what is spoken at each position), showing through co-design that screen reader users want rich, structured exploration rather than a single summary [139]. Olli operationalized part of this insight as an open-source adapter that renders structured traversal for existing charting libraries [14]. A small number of production systems, notably Highcharts, SAS Graphics Accelerator, and Visa Chart Components, have shipped meaningful accessibility functionality [54, 101, 123]; they remain exceptions, and Chapter 4 examines that practitioner tooling landscape in detail.

Across all four modalities the same pattern recurs: each began with converting visual output into another channel, and each has progressively rediscovered that access is not a conversion problem but an interaction problem, turning toward structure, navigation, interactivity, and user control. The research itself has shifted accordingly: where early decades produced perception studies and one-off substitution systems, the last several years have produced co-designed interactive systems, design dimensions, and authoring environments. That convergence is part of why we argue, later in this chapter, that the remaining hard problems are not modality problems at all.

2.2.2 The Split Between Research and Practice

Before turning to the critical literature, it is worth naming that research and practice participate in accessibility at very different speeds, through very different methods, and with very different artifacts.

Accessibility *practice* organizes itself around standards. The Web Content Accessibility Guidelines carry legal and policy weight across much of the world [124], and a profession of accessibility specialists exists largely to translate such guidelines into concrete implementations. One example of this work would be ensuring that interactive elements carry functional semantics that assistive technologies can interpret, which is a foundational, simple, and necessary requirement for accessibility [125].

Standards are powerful precisely because they are consensus artifacts, or *politically mediated documents*, and for the same reason they are inherently reactive: developing, vetting, and formalizing them takes years, so they chronically trail the technologies they govern and are subject to intervention by individuals and organizations. Interactive data visualization sits at a painful point in that lag. Charts are semantically dense, interaction-heavy, and rendered through technologies the guidelines were not written for, so a practitioner who consults the standards in good faith finds little that speaks to their actual artifact. Robust evaluation in practice therefore leans on expert judgment and, in the best cases, direct involvement of people with disabilities, both of which are scarce resources [94].

Technical accessibility and assistive technology *research* meanwhile (outside of the previously mentioned work on modalities and our empirical record), moves quickly and publishes minimally-viable prototypes. These prototypes are an artifact practice largely cannot consume because it requires technical translation and methodological generalization from the small scope of an artifact into a system-wide implementation [31]. A practitioner cannot install a paper; a paper must be read and artifact carefully dissected before being reconstructed or replicated. The result is a conversation in which researchers accumulate evidence and demonstrations on one side, practitioners wait on standards that lag by a decade on the other, and very little crosses between them. This split is not unique to visualization, but visualization makes it unusually visible, and closing it from the tooling side is the project of this thesis.

2.2.3 Empirical Depth, but a Lack of Applied Work in Production

Taking stock of the conversation so far: the field has produced three things in abundance, and two things almost not at all.

What exists, first, is empirical breadth. Surveys and meta-analyses have mapped the research space and its biases: accessibility research broadly concentrates on blindness and on computer output [76], and visualization accessibility research mirrors that concentration, with vision-related work dominating while motor, cognitive, neurological, and vestibular access remain nearly unstudied [70, 83, 132]. Even within the well-studied territory the breadth has a shape: blind and low vision people are routinely researched as one population despite using different assistive technologies and different interaction practices [118], and the prototypes the field celebrates are overwhelmingly built for screen reader interaction in particular.

Second, the field has produced exemplary prototypes. The systems surveyed above, from VoxLens, Chart Reader, Olli, and Umwelt, are rigorous, validated, and often co-designed with disabled collaborators.

What barely exists is the other half of the pipeline: practitioner tools in production environments, and practitioner action at scale. Studies of deployed visualizations and of the people who build them find that accessible outcomes remain rare and that practitioners struggle even when motivated, gathering fragmented guidance themselves with little way to judge its quality [64, 110]. Mainstream visualization tooling has invested its enormous momentum elsewhere, in expressiveness, scale, and interactive speed [15, 50, 102], with accessibility absent from core abstractions and left to individual developers, libraries, or projects. Research prototypes, meanwhile, rarely become infrastructure: toolkit research is evaluated primarily by demonstration [72], and most demonstrated systems are archived with their papers.

This asymmetry is the load-bearing observation of this chapter: **the field has a breadth of empirical work, yet it lacks practitioner tools in production environments and practitioner action.** A practitioner today inherits strong evidence about what good access looks like but material change remains far behind. The natural next question is to investigate why this is the case, and whether and how tool-making can play a role in practitioner action.

2.3 Critical Voices and Work Along the Periphery

The second register of the conversation is critical, and although it is sometimes treated as peripheral to systems research, it sets the terms under which systems research should be read. We treat it here as load-bearing. This section will be brief, but essential.

The disability rights movement crystallized its central demand in the principle "nothing about us without us:" people with disabilities must hold power in the decisions and designs that concern them [20]. HCI and accessibility research have been working through what that demand means for our field's methods, authorship, and self-conception [55, 113, 116, 117], including calls for a crip HCI and crip visualization that engages disability on disabled people's own terms [57, 130] and concrete practices such as crediting disabled participant-contributors by name [138].

Several concepts from this literature directly shape this thesis. Mingus names *access intimacy*, the felt sense of someone genuinely getting your access needs, to insist that access is relational rather than merely technical and logistical [88]; Bennett et al. build on that disability justice lineage to propose *interdependence* as a frame for assistive technology research, countering the field's fixation on independence [8]. Hamraie and Fritsch's crip technoscience manifesto insists that disabled people are experts and designers of everyday life, and re-frames access not as a finished state but as friction, something contested and negotiated [47]. Hamraie's history of universal design develops *access-knowledge*, showing that what counts as access has always been produced through political struggle rather than neutral technical progress [44, 45]. Bennett et al.'s biographical prototypes push against savior narratives in design by recognizing the prototyping disabled people already do in their own lives [9]. Shew names the broader cultural failure *technoableism*: the belief that technology will "fix" disabled people, held mostly by people who never asked disabled people what they want [111]. And within visualization specifically, Hsueh et al. bring crip technoscience to bear on data experiences, articulating how *access friction* arises when one design that includes some people excludes others [57].

We read this literature first as a set of constraints on what this thesis may claim. It is why we reject techno-solutionist framing outright: no tool in this thesis "solves" accessibility, because access is negotiated among people, institutions, and policy, and tools do not meaningfully redistribute power entirely on their own. It is also why access friction appears throughout this thesis as a design reality rather than an edge case, and why personalization is treated in Chapter 8 as an ethical question and not merely an intellectual problem to solve. And this body of work imposes a boundary we want to name early: a thesis about practitioner tooling mostly serves practitioners building *for* disabled users. The later chapters, Cross-perception and Softerware, begin to shift who holds the power to shape our interfaces. The critical literature is the measure of how far there is still to go.

We also read this literature as relatively divorced from accessibility work in practice. Critical scholarship has even been critical of "access" as a framing and its shortcomings [44, 45, 46, 94, 130]. Yet, practitioners are building systems all over the world that must be made accessible. And rather than argue that people with disabilities should not be included in the design and development of new projects, instead we argue that, at least in some cases, the empirical record of what people with disabilities want and need is already enough imperative for builders and designers to begin working. For something new and unexplored? Yes, people with disabilities should be included. But what if we have a sufficient breadth and depth of guidance from people with disabilities already? Aren't we then ready to take action and begin practical application? *Chartability* in Chapter 4 takes this stance: people with disabilities have already said enough for us to build out comprehensive guidelines and an evaluation framework. Between standards, research, and community practice, we have an authoritative set of criteria we can test against. This should be our starting point; the bare minimum. And *Data Navigator* in Chapter 5 didn't seek to build a novel interaction but rather build a better system than what currently exists, following existing empirical work that laid the foundation for rich navigation as a necessity [139]. And in Chapter 5 we *do* demonstrate that our system enables novel interaction designs and how practitioners can approach co-design in a simple, effective, and thoughtful way with partners with disabilities. Chapter 6 sought to investigate the barriers that sighted practitioners (without disabilites) face, which is outside of the scope of critical disability literature, and Chapters 7 and 8 then centered on the experiences of blind and low vision people, and how they might have interactive and innovative power over their experiences.

2.4 Why Navigation, Interaction, and Personalization

If the gap were uniform, any contribution would do. It is not uniform. Looking across the empirical record, the prototypes, and the practitioner studies, the places where practitioners most consistently lack the means to act cluster in three domains, and each domain fails for a different kind of reason.

It is worth being explicit about why these three and not others, because the field's other major needs are of a different kind. Description is the best-served domain: practitioners have models, guidance, and concrete heuristics for what to write [65, 74], and what remains is largely a matter of doing it. Evaluation is hard, but it is hard as a knowledge problem, a matter of gathering, judging, and applying scattered guidance, and a well-built resource can carry a practitioner a long way; Chapter 4 contributes exactly such a resource and uses it to map the territory. *Chartability*, from Chapter 4 ultimately ends up framing the rest of this thesis. As we did work alongside our co-designers, clients, and industry partners over the years, applying *Chartability* to their work, it became clear that three major barriers emerged as the most pernicious (or least-resourced) for practitioners to address.

Navigation, interaction, and personalization are different in kind: in each of them, a practitioner who knows precisely what good access requires still cannot build it, because the means do not exist in their tools. Guidance cannot substitute for means. These areas, then serve as opportunities to innovate, to build better, more general, or more specific tooling.

We unpack these three tooling gaps more below.

2.4.1 Navigation

The empirical case for structured navigation is now strong. Zong et al. showed through co-design that screen reader users want to traverse a visualization's structure, not only hear a summary: structure, navigation operations, and description together determine the quality of the experience [139]. Systems work has begun to deliver this for specific stacks, with Chart Reader designing navigation experiences with screen reader users [121] and Olli adapting existing charting libraries into traversable structure [14]. What practitioners still lack is twofold. They lack low-level building blocks: a way to give *any* data interface, whatever its renderer or framework, a navigable structure with arbitrary shape, rather than the fixed hierarchies particular systems assume. And they lack any visible, manipulable representation of navigation structure in their workflows: a sighted practitioner can see a color palette and act on it with existing tooling but cannot see a navigation graph and act on it, which makes navigation design incredibly burdensome to iterate on and wholly downstream from other visual work. Chapter 5 (*Data Navigator*) contributes the structural substrate, and Chapter 6 (*Skeleton*) contributes the authoring representation; their chapter-level related work sections cover visualization toolkits and authoring systems more in depth.

2.4.2 Interaction

History shows that input design, not just output conversion, changes what is possible. Kane et al.'s *Slide Rule* contributed multi-touch techniques that let blind users operate touchscreens, outperforming a button-based baseline and prefiguring techniques that later appeared in commercial screen readers [66]. That simple lesson has not yet reached data analysis. Accessible visualization interaction today is overwhelmingly what we call *access-oriented*: it exposes information that is already there, through navigation, sonification, or description. These methods are *interactive* as existing work as argued [139], but we argue that the interaction is focused on perception of the representation and not enabling analytical interaction with and manipulation of the information that lies beneath the representation. Analytical work of the kind sighted analysts perform with cross-filtered, coordinated views depends on rapid, continuous loops of input and perception [50, 73], and a screen reader's serial, query-at-a-time interaction may be structurally unable to sustain such loops. Fast, complex manipulation of a data space is presently made inaccessible by the primary assistive tool used by blind analysts. Rethinking input for blind analytical interaction, beyond an *access-oriented* framing, is nearly unexplored territory. Chapter 7 (*Cross-perception*) takes this on, formalizing a design framework and contributing a hardware prototype; that chapter's related work covers tactile and haptic input devices in more detail.

2.4.3 Personalization

Accessibility research has long argued that systems, not users, should do the adapting. Put simply, this means that systems must be capable of adaptation. To me, this is the "soft" part of *software*. Yet, this softness can create conflict in system design, especially when an entire field of work must refactor their dominant tooling in order to enable system adaptation by end users. Visualization faces this exact challenge.

Existing work has demonstrated that interfaces adjusted to fit a person's abilities and preferences measurably improved speed, accuracy, and satisfaction for users with motor impairments compared to default interfaces [33], and ability-based design generalized the stance into principles [133]. Good systems are made to fit a user's abilities and remove barriers that they face. It could be argued, in broad strokes, that the field of HCI centers on the goal of *fit* between a user and a computational system.

Yet, visualization toolkits are largely not made to be personalized by end users. The are rigid and *hard*; lacking that softness required for end-user manipulation. Visualizations are typically built with a *universal design* [45] approach, when being made accessible (if being made accessible at all): one design is intended to fit as many people as possible. In visualization, recent work shows screen reader users benefit from customizing even small things, like the textual tokens spoken during navigation [63]. Yet a significant gap exists that enables visualization systems to adapt to personalization at scale and for many different people. How should these systems be built and what would they require? Which dimensions and features and functionality of data visualizations *should* be manipulable and customizable to fit end users? How should systems be constructed in order to make personalization effective and useful? And ultimately: how do we make visualization toolkits *softer*?

Some existing work on customizable accessibility settings identifies what drives people to adopt or abandon personalization features [135], but not all of this work neatly translates into visualization system design and development. Additionally, and fundamentally, we simply lack an empirical record of what exactly different people with disabilities would even want or need to manipulate. Presently, no mainstream visualization library or system offers practitioners infrastructure for end-user preferences over data representations: no shared option vocabulary, no persistence, no way to reconcile preferences with authorial defaults. Most visualizations become completely static in their design. Those with manipulable elements are only manipulable through hacking or manual overrides, placing a significant burden of access on end users, exactly where ability-based design says it should not [133]. Chapter 8 (*Towards Softerware*) engages this domain in collaboration with a production charting library, considered by the empirical record to be the most-accessible toolkit according to present evaluation methods by a wide margin [71]; Chapter 8's related work section covers adaptive systems and personalization research in more depth.

2.5 What to Take Away

Five things from this chapter frame everything that follows. First, tools direct the work and the workers who use them, which makes tool-making a rich potential site for accessibility intervention and research into better understanding practitioner barriers when making and using tools. Second, the conversation between accessibility and data visualization is old, rich, and increasingly self-critical. Third, the field's output is lopsided: empirical breadth, exemplary prototypes, and guidance exist in abundance, while production tooling and practitioner action barely exist. The broader field's critical voices impose real constraints: tools do not solve access, and this thesis will not claim they do. Instead, critical work implores us to bridge the gap between research and practice. People with disabilities have been saying enough, it's time to get to work and try to apply what we know at scale. Fourth, the gap is concentrated where practitioners' means run out in three specific ways: navigation lacks a structural substrate, interaction lacks input techniques for analysis, and personalization lacks infrastructure. Fifth, the first step in our thesis work should be to collect the empirical record and synthesize it for applied accessibility work in the context of interactive data visualization. This is Chapter 4's *Chartability* project. Given where the field stands, *navigation*, *input*, and *personalization* are the categories of work that follow Chapter 4; the chapters ahead carry this out as the body of this thesis, and our concluding chapters return to what tool-making accomplished and what it cannot accomplish, on the other side.

Chapter 3
Overview of Contributions

In this dissertation, we contribute practical advancements in tool-making as an intervention on the accessibility of interactive data experiences. The thesis of this dissertation is as follows:

This dissertation argues that the tools practitioners use to build interactive data experiences are themselves sites where accessibility barriers are produced, prevented, or alleviated for both end users and authors. This work contributes five tools, Chartability, Data Navigator, Softerware, Cross-perception, and Skeleton, that collectively center accessibility work on the empowerment of disabled and non-disabled practitioners across the full arc of evaluation, data navigation, analytical interaction, and personalization.

Rather than framing accessibility research solely around ideal experiences for end users with disabilities, this thesis investigates why accessibility work is so difficult for the practitioners who build interactive data experiences and what tool-making can reveal about those difficulties. We organize this investigation around four domains where practitioners face the most persistent challenges: evaluation, navigation, interaction, and personalization. *Chartability*, a heuristic framework contributed first and maps the full landscape of accessibility barriers. *Chartability* helped us to identify and prioritize our three latter domains (navigation, interaction, and personalization) as the areas where the most severe gaps remain in practitioner work. *Data Navigator* and *Skeleton* then investigate navigation, finding that practitioners struggle because navigation structure has no visible, manipulable representation in their workflows. *Cross-perception* engages interaction, demonstrating that blind data analysis has been constrained by existing tools and that a new interaction design framework can reshape what analytical work is possible. *Softerware* addresses personalization, revealing that access needs genuinely conflict across users and that meaningful personalization requires system-level infrastructure that does not yet exist in practitioner tooling ecosytems.

We engage each domain below with the questions: “Why does this work matter?”, “Why is it hard?”, and “What has tool-making within this domain showed us?”

The four domains of work that we engage in this thesis start first with **evaluation**. In work contexts where someone is designing and developing interactive data experiences, the practitioner must have the knowledge, tools, and resources available to systematically identify how their interfaces produce barriers for people with disabilities. A significant portion of professional accessibility work (arguably most, if not all) is founded on auditing and evaluating barriers to access. This work is pre-dominantly done through a standards-based approach [124], although in more robust evaluation work, people with disabilities are actively involved in the process [94].

Evaluation is difficult work because much of it is contextually defined by the author themselves, and most tasks at this intersection require careful, non-automated processes and methods [25, 94]. To make matters more difficult, no comprehensive guidelines, tests, and tools exist in any singular location. Practitioners often must gather these resources themselves, which tend to be situated towards accessibility in general or are high-level and provide minimal usefulness in practice. Additionally, practitioners themselves often have little knowledge about the veracity or quality of any given bit of information they gather [64, 110], and often do the work themselves to synthesize this disparate space of information into a usable format they can apply to their own evaluation work.

To engage this, our first main chapter focuses on *Chartability*, a heuristic framework that enables designers, developers, and auditors to systematically evaluate data visualizations and interfaces for a wide range of accessibility barriers, considering people with visual, motor, vestibular, neurological, and cognitive disabilities. In this project, we did the hard work for other practitioners and contributed our collection of synthesized accessibility resources in a single workbook. We had practitioners try out our resource in real environments, in-situ, in order to learn more about the challenges and barriers they themselves faced in evaluation work. With *Chartability*, practitioners, especially those with limited accessibility expertise, gained more confidence and clarity in assessing and improving their work. Additionally, *Chartability* has since become widely applied as a framework that isn't just used for evaluation but also as design guidance in many contexts, internationally, including policy organizations, governmental groups, and more than 100 companies and businesses.

Chartability then opened up a significant landscape of new projects and research directions. From a combination of my existing expertise as a visualization designer and engineer, in addition to our continued application of *Chartability* in the wild, we began to identify the trickiest and most-difficult domains of work for practitioners. Chartability has 50 total heuristics, or tests, each organized under one of seven principles. These three domains did not simply "emerge" on their own; they were evidenced by our applied work. Alongside the research activities that comprise this thesis, in a freelancing role we applied *Chartability* by partnering with industry, government, and non-profit organizations, conducting over 150,000 tests (both manual and automated) and working closely with dozens of designers and engineers. From that body of work, three larger domains stood out as the areas where the most severe and dramatic accessibility barriers remained unaddressed: data *navigation*, analytical *interaction*, and interface *personalization*.

So the next section of this thesis engages the first of these three: **navigation**. Navigation is a fundamental type of interaction that is leveraged by modern software-based assistive technologies. Screen readers, the primary tool used by people who are blind to interact with computers, navigate content. Additionally, many other assistive technologies, such as a sip and puff device (like the "POSSUM" from as far back as '63 [79]) also navigate. Navigational technologies are leveraged by people with a significant array of disabilities, yet tend to be entirely ignored by existing data visualization tools, which are pre-dominantly built to support direct input (using a computer mouse).

Empirical work has already demonstrated that structural navigation is actually good [139], even when regular alternative text (image descriptions) exist. This is both because people who are blind can gain both a high level understanding (from the description) as well as lower-level sense of the data's structure and arrangement, in addition to the fact that discrete, structural navigation exposes interactivity that may exist on any visualization elements (such as they can be hovered or clicked with a mouse in order to perform some action). So, if good empirical work exists: *why* *haven't practitioners put this research into action? What makes this work hard to do?*

We first built *Data Navigator* to provide the building blocks we needed in order to address the technical and conceptual gaps that were required to make any visualization or visualization tool provide a navigable, interactive structure. *Data Navigator* is a low-level toolkit which can be used to construct accessible navigation structures such as lists, trees, and diagrams from an underlying graph structure. We leveraged graph theory for an applied HCI problem: nodes and edges represent any relationships within the data as a structure, which then supports rich expressiveness of data navigation experiences. Users can navigate discrete marks in a visualization, clusters, groupings, and more. In addition to its structural scaffolding, *Data Navigator* also supports a wide array of input modalities leveraged by people with disabilities (screen readers, keyboards, speech, and gestures).

Data Navigator provided a substrate, but this contribution alone wasn't enough to engage the question *why is navigation so hard, in practice?*. So the chapter following *Data Navigator* introduces *Skeleton*, a data navigation authoring tool built on top of *Data Navigator*. Our novel approach in *Skeleton* involves visualizing and making manipulable the nodes, edges, and textual data that comprise non-visual end user experiences. *Skeleton* visualizes the building blocks that comprise *Data Navigator*. Additionally, *Skeleton* provides expressive, rapid scaffolding capabilities that leverage data visualization rendering engines. This scaffolding engine helps practitioners quickly create common configurations for non-visual data navigation structures that retain visual congruence to the underlying structure.

But most importantly, *Skeleton* serves as a framework that shapes designerly consideration. Our conjecture was that because sighted practitioners cannot *see* navigation building blocks, they will not treat those elements as iterable design materials. We conjectured: Navigation is hard in practice because sighted designers face barriers to iteration and understanding. We conducted an empirical study with sighted practitioners and found that making non-visual elements visual helped practitioners shift from treating accessibility as a compliance task to treating it as a design problem, re-iterating on the visual aspects of their design, and engaging in the complex and nuanced components that comprise data navigation experiences.

Now, **interaction** becomes the next area we want to engage. Existing accessible data interaction for people who are blind, including our previous work on navigation (which is a form of interaction), predominantly seeks to expose information. This is what we call *access-oriented interaction*. In terms of low-level analytical tasks, most are then made feasible through navigation, sonification, or summarization-based and question-answering approaches: retrieving values, filtering, computing derived values, sorting, determining ranges, clustering, and finding outliers. What remains are analytical tasks that, despite being "low level" (understood as *unable to be reduced into more fundamental tasks*), are cognitively highly complex: finding correlations and characterizing distributions [3]. These tasks require complex hypothesization and exploration, rather than a system that simply encourages surfacing what is known or what is already present in the data: it requires combining, remixing, restructuring, and dividing data.

But blind *analytical interaction* isn't just important to engage because it is understudied, it is important to engage because many of interactive information visualization's most impactful tools for data science enable it [32, 50, 91, 127]. In visualization, *cross-filtering* is one example of an interaction that enables a user to filter one visual space while seeing a coordinated change in another visual space simultaneously and near-instantaneously. The speed of input interaction and perception of output also matters: even a small bit of latency changes the quality of a user's data exploration activities [73]. We conjectured that a screen reader, the most-used tool leveraged by blind people when interacting with computers, may be insufficient for engaging this task.

To engage this, this thesis introduces *Cross-perception*, an approach for building analytical interactions that support perception in one space of input interaction with simultaneous, non-competing perception of output in another space of data representation. We first formalized a design framework for producing *cross-perception* experiences and then built a novel prototype device, the *cross-feelter*, that enables blind *cross-perception* of a cross-filtering data exploration interface. In an empirical study with blind users (with and without existing data expertise), we found *cross-perception* speeds up analytical exploration by 90% and helps blind users consider vastly more questions of their dataset (+188% computational queries, +54% spoken aloud) compared to a screen reader-driven interaction.

Beyond performance, we found that our input modality itself shaped the character of analytical engagement: participants didn't just work faster, they considered more dimensions of their data and asked qualitatively different questions. The *cross-feelter* also reduced anxiety and substantially increased enjoyment, particularly for participants without prior data expertise, suggesting that the barriers blind practitioners face in data work are not only functional but affective. Additionally, we had our blind practitioners imagine new interaction possibilities that *cross-perception* could enable including and beyond our *cross-feelter* device.

Our final domain of work is the most difficult for visualization practitioners to engage: **personalization**. While *navigation* demands better software tooling and visual support and *interaction* requires new hardware, *personalization* completely re-orients how software authoring takes place. Personalization matters because of *access friction*, which is a design challenge where one design or interface configuration that might be accessible for one person or group of people turns out to create barriers for someone else [44, 57]. In existing work on personalization and accessibility, studies have demonstrated that end-user control is great to have and can alleviate friction [63, 135], but little work has been done to explore what personalization looks like for an existing data visualization library and how practitioners should build and maintain their existing systems to support it.

Our final chapter introduces *Softerware* to address the tension between standardized accessible design and the diverse needs of real users with disabilities. In the wild, *access friction* exists in every design that reaches a public audience; it is inevitable. This tension ultimately means that some users have a worse experience, and may even face exclusion, with any particular design configuration. So for this work, we conducted our research in-situ with visualization software engineers and designers and worked to build a scalable, flexible software system dubbed "softerware" that enables end users to manipulate the appearance and functionality of the charts and graphs they encounter according to their own preferences. We conducted empirical research to inform our collaborators, as well as other visualization system authors, with guidelines and considerations for building *softerware* systems. In our study, no two participants chose the same preference configuration and participants with the same diagnosed condition sometimes needed opposite design treatments. Practitioners, meanwhile, immediately raised ethical concerns about whether personalization would let designers off the hook for poor defaults. These findings revealed that the real barriers to personalization are not at the level of any individual chart but at the level of system infrastructure: without persistence, cross-system interoperability, and shared standards, the effort required of end users exceeds the value they receive.

Combined, these contributions provide empirical insights and practical advancements in the state of the art for tooling that bridges gaps in current accessibility practices in visualization and data science. Our work ultimately enables people with and without disabilities to better evaluate barriers in, analyze with, design for, develop, and personalize interactive data experiences. We demonstrate that tool-making is a productive intervention that both engages accessibility barriers and elucidates why those gaps exist in practitioner work.

Part VI

Conclusion

Chapter 9
Findings

The five projects in this thesis were presented as five chapters, each with its own results and its own conclusion. Read separately, they contribute a heuristic resource for evaluation, a structural toolkit for navigation, an authoring environment for non-visual design, an interaction technique and device for blind analysis, and a system and study of end-user personalization. This chapter states what they show *together*. Chapter 2 closed on a lopsided field: a breadth of empirical work and exemplary prototypes on one side, and a near absence of practitioner tools in production environments and practitioner action on the other, with the hardest remaining challenges concentrated in navigation, interaction, and personalization. The findings below are what this thesis learned by working inside that gap.

Each finding is a synthesis across chapters, and each is traceable to results reported in the body of this thesis. We have written them for two audiences at once: practitioners who build, evaluate, and maintain data experiences, and researchers who study how that work happens. Where a promising idea outran the evidence, it is not here; it is in the next chapter, with the rest of the discussion and future work.

9.1 Access Barriers Are Produced for Practitioners, Both With and Without Disabilities

The social model holds that disability is not located in a body but produced in the interaction between a person and a human-made system [82]. This thesis began by applying that lens to end users: visualizations, like curbs, produce barriers. The cross-cutting finding of this thesis is that the same lens applies, with the same force, to the practitioners doing accessibility work, and that the materials of that work are among the systems producing the barriers.

Every project surfaced a version of this. In *Chartability*, the guideline landscape itself was a barrier: standards and research were scattered, often high-level, and difficult to apply to complex interactive data interfaces, and practitioners had little way to judge the quality of what they gathered [64, 110]. Guidelines written to remove barriers for end users limited what practitioners could reasonably understand and act on, and an expert visualization practitioner in our study admitted he did not know where to start. In *Skeleton*'s co-design foundations, *Data Navigator* made better end-user experiences possible, but because its structures were entirely non-visual and code-bound, it produced severe iteration barriers for sighted practitioners: they could not perceive what they were building, and prototypes were misinterpreted or abandoned. And in *Cross-perception*, the screen reader, the most important enabling technology for blind computer use, also limited what blind practitioners could reasonably do analytically: its serial, discrete traversal kept participants in an access-oriented mode that structurally separated interaction from information, with measurable costs in time, completion, and the number of questions they could ask of their data [108].

The operationalized version of this finding: when accessibility work stalls, look for barriers in the practitioner's tools and materials before attributing the failure to motivation or knowledge. None of the practitioners in these studies lacked care. What they lacked, in each case, was a working relationship with materials that had been shaped without them in mind, and this is true whether the practitioner is sighted, blind, disabled, or not. The same observational stance the field has learned to take toward end users belongs in the study of the workers, too.

9.2 Tools Can Reduce the Work Required to Make Things Accessible

Chapter 2 argued that guidance cannot substitute for means. The constructive half of that argument is that supplying means measurably changes what practitioners can do, and three chapters provide the evidence.

Chartability reduced the perceived difficulty of evaluation and increased the confidence of practitioners new to accessibility, whose audit results were reasonably comparable to the authors' own; beyond the study, it has been applied by policy and governmental organizations and more than 100 companies, across toolchains from Tableau and Excel to D3 and Figma. *Data Navigator* gave previously inaccessible rendering formats a navigable structure: a static raster image gained full multi-input navigation with no automatically detectable conformance failures, and a deterministic ingestion pipeline recreated, and then improved on, an existing toolkit's navigation scheme for rendering modes that previously had none. *Skeleton*'s scaffolding collapsed the spatial authoring of a navigation structure to one or two minutes of refinement, and within single sessions every participant iterated substantively on designs that, in a code-only workflow, our co-design collaborators had struggled to iterate on at all.

None of this is evidence that tools solve access; the boundaries of that claim get their own finding below. It is evidence of something narrower and more useful: the labor of access work is sensitive to tooling, and reductions in that labor are achievable, measurable, and repeatable across very different kinds of tools, from a workbook of heuristics to a rendering-independent toolkit to an authoring interface.

9.6 The Hardest Barriers Required Intervention at the Substrate

Chapter 2 characterized navigation, interaction, and personalization as a missing structure, a missing technique, and a missing infrastructure. Having now worked in all three domains, we can state what they share: in each, the barrier sat one level below where practitioners are ordinarily able to act, in the substrate their work is built on, and in each, outcomes changed when the intervention went to that level rather than adding features above it. This finding is held heavily in tension with existing research that tends towards higher levels of software and interaction abstraction, not lower. A massive wave of new research has emerged that essentially asks, "what new layers or wrappers can we create around existing interfaces and user experience problems, using large language models and agents?" While our thesis doesn't engage the effectiveness or intellectual value of those projects, instead we hold this thesis in fundamental contrast to their approach: that some of our most difficult gaps are best engaged below our present levels of software abstraction, not above.

The evidence is consistent across the three parts of this thesis. No amount of feature work on top of canvas-rendered pixels can recover semantics that the rendering substrate never carried, and forcing complex, non-hierarchical data relationships into HTML's hierarchical substrate limited what was possible to build. This is precisely what a large and passionate resistance to "overlay" technologies engages within both the practical and academic accessibility communities [28, 40, 48, 78]. There may simply be a limit to how much good data, if at all, can be generated from bad or missing data [13, 93].

Data Navigator changed outcomes by supplying a structural substrate of its own, a graph independent of any renderer. This requires underlying data that is supplied with sufficient quality, what Doug Schepers calls an "underlay" rather than an overlay [104].

No software feature on existing consumer hardware approximates persistent physical state and simultaneous perception of input and output, the properties analytical interaction depends on [73]; the *cross-feelter* changed the hardware substrate, and the character of blind analysis changed with it, from accessing data toward interrogating it, at nearly three times the query rate.

And per-chart personalization features fail the effort-to-outcome test that end users themselves applied; what every participant and practitioner in the *Softerware* study named instead were system-class needs [31]: persistence, profile sharing, interoperability, and standards beneath any individual chart.

This is why the substrate matters as a space of intervention: in all three domains, the ceiling on what a motivated, well-informed practitioner could make accessible was set by material beneath their reach, so guidance, effort, and even exemplary prototypes hit that ceiling without moving it. The work in this thesis moved outcomes exactly where it changed the material itself, and the same evidence shows the stages of this work leaning on one another: structure without authoring support went unused, authoring raised evaluative questions practitioners could not answer alone, and personalization without accessible defaults and persistent infrastructure shifts burden onto the people it is meant to serve. Researchers and toolmakers deciding where to spend effort in these three domains have, in this thesis, a consistent answer about where the leverage was found.

Chapter 10
Discussion & Future Work

10.2 "Applied" accessibility work and *low research*

I came into an academic research environment already as an engineer. I had questions to big problems, especially about how tools might shape human behavior and how we can use that towards ethically good outcomes. But *Chartability* was a project I made first for myself, to engage the problems I, personally, was facing.

And *Data Navigator* was motivated by my existing experience making visualizations navigable. I knew we had to build better tools, because continually trying to make a hierarchical substrate like HTML work for complex, non-hierarchical relationships simply limited what was possible and easy to do.

These two projects then led to many opportunities to test them in applied contexts. Our CZI grant was motivated towards the application of both, Adobe was interested in *Data Navigator*, and the University of Wisconsin's GIS folks wanted to figure out how to make a complex map navigable, *Data Navigator* or not. Yet, as a researcher I wasn't incentivized to pursue these projects. In fact, they all proved to have no immediately apparent novelty. It was a risk to spend my time in these spaces.

Applied work is structurally and socially devalued in academic research environments, seen often as a lower form of research and design compared to more "pure" forms of knowledge production. (We conjecture that in part this is due to the scope of knowledge that practitioners produce: they work to take broad knowledge and apply it to specific, small problem spaces. Foundational research tends to assume that broad knowledge does not then bubble up from this work, that it is a one-way flow.)

In my first year as a PhD student I was given the advice by a senior researcher not to hold my experience as a practitioner too highly because, "practitioners often don't know what they are doing." But I did, of course, to some degree. Yet, this attitude is pervasive in some spaces and is a problem because it limits what kinds of knowledge we care to attend to.

Initially, I felt that with *Chartability* I had lucked out. But after four more collaborations which resulted in funding and two research publications, we can now argue that this method has been reliable. In *Softerware* and *Skeleton*, pursuing practical problems in applied environments *did* serve as a fruitful source of generalizable knowledge and advancements in the state of the art. But these projects, in addition to *Chartability*, were all motivated by immediate, non-novel problems: people knew their charts were inaccessible and wanted some design guidance or engineering solutions.

So perhaps "low research" works.

Yet, some research communities, especially *ASSETS*, have yet to reckon with "applied" work like this. If you trace the lineage of *action research* from Hayes [49] (which is how we frame our applied work in *Skeleton*), it leads into fantastic areas of participatory and community-based work, often with or by marginalized people and people with disabilities [8, 57, 75, 80, 113, 116, 117, 122, 131]. We would even argue that this lineage of work, which prioritizes and centers the lived experience of a person, is also why we have seen so many successful auto-biographical and auto-innovation styles of research as well recently [12, 34, 35, 55, 106, 129, 130].

And yet, action research largely does not touch on how practitioners (particularly those without disabilities) do applied work in accessibility [64, 110]. We believe that significant gaps exist in the space between traditional "high" knowledge production and applied, "lowly" spaces. Researchers tend to pursue novelty and the lifecycle of research production is highly risk adverse. And yet, relatively little work has been done within the accessibility research community to intellectually address the actual social and cultural risks involved producing decades of useless, lost, and forgotten prototypes and design guidelines [58, 60]. We would conjecture that research-producing academics are overly concerned with academic readership relative to the myriad of ways that their knowledge production actually is used by the rest of the world [89]. We would challenge *ASSETS* and the AT/accessibility communities of *CHI* in particular to take seriously the lack of applied (industry, government, and non-academic) work it produces as a community.

10.3 Who is responsible for repair?

Lastly, we want to revisit one of this thesis's opening points, where we argue that the *tool-makers* are first responsible for repair. This is true. However, the most pressing issue we have faced in recent years is mostly unmentioned across these research projects: tool-makers might be responsible, but this is because they are the only ones who have the *power* to make things accessible. Does this always need to be the case? Can we imagine an artifact's authority over the user's ability to access being designed towards self-subversion [38] or de-centralized agency [19, 68, 90], instead? What might that look like, concretely?

In *Softerware*, we begin to engage this larger problem in terms of an idealized state where a user can repair or re-design their own experiences. But to me, this self-repair is like laying down train tracks for yourself as you move a locomotive, but then lifting up your own tracks behind you as you go. You're the only one helping yourself. This is not ideal, for you or others.

What we need are broad, lasting, infrastructural changes. On the web, this problem becomes quite difficult to solve. A personal computer or device? Again, someone can auto-design their interfaces into a better state. But back when we started *Chartability*, the WebAim Million's report showed more than 95% of the top one million website home pages contain at least one critical accessibility error. And now, more than six years later, that proportion is unchanged [128].

Some had imagined that generative AI would solve the massive infrastructural repair problems we face. But unfortunately, the latest WebAim Million report shows that since 2020, ARIA usage has increased and correlates to more errors, while use of tabindex on a page has increased nearly 300% and also correlates to more errors on a page. If anything, during the age of generative AI, we have seen existing bad patterns worsen in prevalence and complexity.

We firmly believe that a tools-based approach is not enough on its own. Tool-making cannot be the *only* intervention on inaccessibility. Tools and tool-making, as our thesis argues, have a powerful role to play. But we simply can't tool our way out of failed infrastructure and inadequate policy when someone else *owns* the tools and tool-making. Visiting a website is like going into someone else's home: arranged according to their effort, tastes, and so on. If you can't access their home, you essentially need to request that they let you in personally. Website repair always falls to the owner and maintainer of a website, and they largely don't take any meaningful action.

Sidewalks outside of homes are a good parallel to this problem. Sidewalk accessibility is a massive infrastructural problem [98], and yet localities treat sidewalk maintenance in different ways: some, like where I, Frank, presently live in the south hills of Pittsburgh, put the onus on the homeowner whose house and property the sidewalk touches. In other places, sidewalks are considered a public path, similar to a roadway, and are maintained through public tax and resource management. To no surprise, privately-maintained, public-access sidewalks are worse for people in pretty much every way than publicly-maintained ones [134]. This is because private homeowners don't care about sidewalk maintenance unless the city manages to fine them or they get sued.

And the web is a collection of private spaces that you visit privately. There is no truly shared, universally democratic, public space on the web. Centralization is partly to blame: sharing space while scaling leads to consolidation.

So our future work will continue to wrestle with the same tensions of scale, repair, and anti-consolidation of power, motivated by the same WebAim Million report. But now we look to questions of *democratic* and *radical* access to accessibility repair. The barriers we hope to tackle in the future are political and infrastructural. Perhaps tool-making will participate in this work, but it seems clear now from our work that the upstream technical problems and socio-political conditions that tools inherit, will likely not be addressed by tools alone.

What does "democratic" and "radical" infrastructure work look like? It will probably be an extension of *Softerware*, to some degree. We imagine future research that explores public-first spaces, ones where access is socially negotiated and repair belongs to all of us. Is this an autonomous space, like an autonomous zone [11] separate from the web? Above it, looking down into it, like shared annotation tools but capable of sharing the manipulation of websites [97]? A space with ambient co-repair, modeled after projects that bring people together [107] or that allow community "fixing" of misinformation [61]? Perhaps feminist thought on the ethics of care can help us [51, 77]? Or maybe it will be something else entirely; I'm not yet sure. But what made the web fantastic years ago is long gone; most of it has been hedged into corporate spaces that are controlled, maintained, and repaired by corporate power. And these entities are notoriously bad at repair. What we imagine in the future involves reclaiming a sense that the web is *ours*, belongs to *us*, and that ultimately *we* are responsible for making it accessible.

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